Open-Source Predictive Maintenance System for AC Induction Motors Using LSTM Neural Networks and Low-Cost Embedded Hardware

Juan Pedro de León Sum
Instituto Tecnológico Regional Norte
Universidad Tecnológica (UTEC)
Rivera, Uruguay
juan.deleon@utec.edu.uy

Eder Mateus Gonçalves
Universidade Federal do Rio Grande (FURG)
Rio Grande, Brazil
edergoncalves@furg.br

***Abstract*—This paper presents the design, implementation, and proof-of-concept laboratory validation of a fully open-source, low-cost condition monitoring system for AC induction motors based on LSTM forecasting of vibration signals. The platform integrates wireless sensing nodes (ESP8266, MPU-6050, and LM35; bill of materials below USD 15 per node) with a Raspberry Pi gateway and a centralized LSTM predictive engine. Experiments on a single-phase and a three-phase induction motor, under static imbalance and a controlled progressive-loading protocol, show that a single-layer LSTM with 4 385 parameters reaches a similarity score of $S = \mathbf{95.30\,\%}$ and a correlation of $r = 0.82$ on a held-out subset, although point-wise accuracy on a larger held-out subset remains limited ($R^2 \approx 0$). Applied autoregressively to an independent experiment, the model indicatively projects the crossing of the ISO 20816-1 Zone A/B reference threshold approximately 1 h 32 min after the end of the seed sequence.**

***Index Terms*—predictive maintenance, LSTM, induction motor, vibration analysis, ESP8266, open-source, ISO 20816, Industry 4.0.**

I. Introduction

Electric motor-driven systems account for an estimated 43 % to 46 % of global electricity consumption, and industry is their largest end user [1]. In agro-industrial settings, unplanned motor failures during critical production seasons can result in significant economic losses. Despite these risks, many small and medium-sized enterprises (SMEs) in developing regions continue to rely on corrective or time-based preventive maintenance, which does not leverage the continuous data generated by operational equipment.

Predictive maintenance (PdM) based on condition monitoring can reduce unplanned downtime by 30 % to 50 % and maintenance costs by 10 % to 25 % compared to reactive strategies [3]. However, adoption is often hindered by commercial solutions that involve recurring subscription fees, proprietary data silos, and limited access to historical data [4], [5]. These factors are significant barriers for enterprises seeking technological sovereignty and scalable monitoring.

The convergence of affordable microcontrollers (e.g., ESP8266/ESP32), open-source machine learning (ML) frameworks (e.g., TensorFlow/Keras), and open databases (e.g., MariaDB/InfluxDB) offers an opportunity to build accessible PdM systems [6]. Long Short-Term Memory (LSTM) networks are well suited to vibration-based degradation forecasting because they model the temporal dependencies inherent in progressive mechanical degradation [7], [8]. Nevertheless, most published implementations rely on proprietary hardware or do not release their full hardware and software stack, leaving a gap for low-cost, reproducible, end-to-end systems aimed at resource-constrained industrial contexts.

This paper addresses this gap with a fully open-source, end-to-end PdM system. The key contributions are: (1) a low-cost open hardware design (bill of materials below USD 15 per node) using locally sourced components; (2) a wireless Raspberry Pi-based gateway architecture; (3) an LSTM hyperparameter search, evaluated with both a distribution-level similarity score and point-wise error metrics; (4) a vibration-velocity processing chain and an autoregressive forecasting procedure evaluated against ISO 20816-1-referenced alarm thresholds; and (5) the full public release of all hardware designs, firmware, and software [13].

II. Background and Related Work

Scalabrini Sampaio et al. [9] reported up to 93 % accuracy using ANN, regression tree, random forest, and SVM models for vibration-based motor failure prediction, relying on more extensive feature engineering than the variance-based approach used here. Eang and Lee [10] reported accuracies above 92 % with CNN-RNN observers for fault detection in DC motor drives of industrial robots. In previous work, the author implemented TinyML fault classification on an ESP32 with an ADXL345 accelerometer, reaching accuracies above 92 % [11]; that system, however, was limited to binary fault/no-fault classification without degradation forecasting.

For time-series degradation forecasting, Hochreiter and Schmidhuber [7] introduced the LSTM architecture to address the vanishing-gradient problem in recurrent networks, enabling the learning of long-range temporal dependencies, a property suited to the progressive degradation patterns of rotating machinery. Reference implementations of LSTM-based PdM

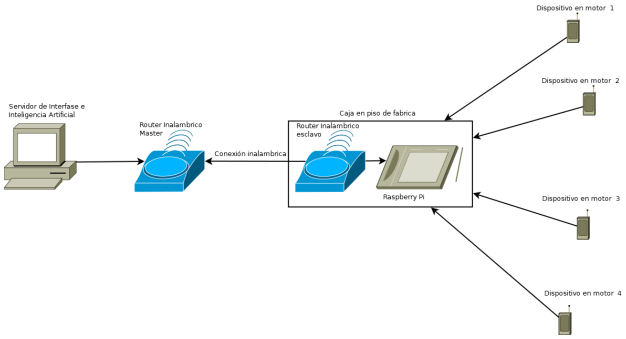


Fig. 1. Three-tier system architecture. Sensing nodes transmit via Wi-Fi to the Raspberry Pi gateway, which stores data in MariaDB and forwards them to the LSTM server.

[8] show that compact recurrent models can be effective on structured industrial time series.

The gap addressed here is the absence of a system that simultaneously offers: degradation-trend forecasting, very low hardware cost (below USD 50 per node), full open-source availability, multi-hour degradation forecasting, and alarm thresholds referenced to ISO 20816-1. Validation of the present work is performed in a laboratory setting; validation in an operational plant is left for future work.

III. System Design

A. Three-Tier Architecture

The system follows the three-tier architecture shown in Fig. 1: (1) distributed sensing nodes attached to each monitored motor; (2) a Raspberry Pi plant-floor gateway acting as wireless access point and local database; and (3) a centralized x86 server hosting the LSTM inference engine and Grafana/InfluxDB dashboards. All tiers communicate wirelessly, requiring no cabling modifications on the plant floor.

B. Sensing Node

Each sensing node integrates: (i) an ESP8266 Wemos D1 Mini (80 MHz Tensilica L106, 4 MB flash, Wi-Fi 802.11 b/g/n) as processing and communication core; (ii) an InvenSense MPU-6050 triaxial MEMS accelerometer on a GY-521 breakout board, mounted on the carrier board beneath the microcontroller (Fig. 2b) ($\pm 2\,g$ full scale, I^{2}C bus at 400 kHz, 16 384 LSB/g) for vibration acquisition; and (iii) a Texas Instruments LM35 analog temperature sensor (10 mV/°C output, ±0.5 °Caccuracy, −55 °C to 150 °C range) for thermal monitoring. Each node is identified by its ESP8266 MAC address, enabling plug-and-play addition without manual configuration. The electrical schematic was designed in KiCad 6.0 (GNU GPL v3), and the node is housed in a 3D-printed PLA enclosure bolted to the motor frame through an aluminum bracket close to the mounting feet (Fig. 2), with power supplied over USB. This mounting point differs from the bearing housings recommended by ISO 20816-1 [2], and the enclosure and bracket form part of the mechanical transmission path to the sensor. The bill of materials is below USD 15 per node, with all components locally sourced in Uruguay. Schematics, PCB layout, and enclosure files are available in the project repository [13].

C. Plant-Floor Gateway

The gateway (Fig. 2c) combines a consumer-grade Wi-Fi router board, removed from its housing and installed inside the gateway enclosure, which provides the wireless access point and DHCP service for the sensing nodes, with a Raspberry Pi 3 Model B running Raspberry Pi OS Lite (headless), connected to the router over Ethernet. This arrangement requires no plant cabling modifications. Data received from the nodes are processed by a PHP script on an Apache web server and stored in MariaDB (fields: node MAC address, triaxial acceleration variance $\sigma^2_{a,x}$, $\sigma^2_{a,y}$, $\sigma^2_{a,z}$, integrated vibration velocity v_{RMS}, surface temperature in °C, and UTC timestamp). Both boards are housed in a sealed enclosure with a 12 V switched-mode power supply, a voltage-regulator board, and a cooling fan.

D. Signal Acquisition and Processing

The MPU-6050 accelerometer was operated at its native output data rate, $f_s = 1\,\mathrm{kHz}$, using the I^{2}C fast mode (400 kHz) of the ESP8266. Each acquisition cycle nominally lasts 1 s: a 0.5 s sampling window ($N = 500$ acceleration samples) followed by a 0.5 s interval for processing and Wi-Fi transmission. This yields one feature record per second (4 500 records in a 75 min run). The model input is the X-axis acceleration variance, denoted σ^2_a in the following; the X axis was oriented radially, parallel to the motor base, and showed the highest sensitivity to imbalance in Protocol 1.

For each block, a digital high-pass filter with cutoff $f_c = 10\,\mathrm{Hz}$ removes the static gravity component and low-frequency drift. The filtered acceleration $a[k]$ is integrated with the trapezoidal rule, $v[k] = v[k-1] + \frac{\Delta t}{2}\,(a[k] + a[k-1])$, with $\Delta t = 1/f_s = 1\,\mathrm{ms}$, and the vibration velocity is summarized per block as $v_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} v[k]^2}$. The acceleration variance σ^2_a of the same filtered block is the quantity used as LSTM input.

The internal digital low-pass filter (DLPF) of the MPU-6050 was set to its widest configuration (DLPF_CFG = 0), with a nominal accelerometer bandwidth of 260 Hz, well below the 500 Hz Nyquist frequency; this attenuates components that would otherwise alias into the band of interest, although, since the filter acts in the digital domain, residual aliasing from strong components above 500 Hz cannot be fully excluded. Combined with the 10 Hz high-pass filter, the effective measurement band is 10 Hz to 260 Hz. ISO 20816-1 specifies 10 Hz to 1 000 Hz for machines above 600 r/min. The reduced band covers up to the 5th rotational harmonic of the single-phase motor (47.3 Hz at 2 840 r/min) and up to the 10th of the three-phase motor (24.2 Hz at 1 450 r/min), which carry the main signatures of imbalance, misalignment, and mechanical looseness. Because integration weights each spectral component by $1/(2\pi f)$, the omitted 260 Hz to 1 000 Hz band contributes little to v_{RMS} when vibration is dominated by low-order rotational harmonics, as in the conditions tested here; it does, however, exclude the high-frequency content of

early-stage rolling-bearing defects. The resulting v_{RMS} values are therefore band-limited estimates, and the ISO 20816-1 zone boundaries are used as reference alarm thresholds rather than as standard-compliant severity measurements. Ongoing work migrates the node to an ESP32 with an ADXL355 accelerometer to cover the full 10 Hz to 1 000 Hz band.

E. LSTM Model Development

Models were developed in Python 3.7 using TensorFlow 2.x and the Keras Sequential API. The 4 500 records of Experiment 1 were split chronologically into training (70 %, records 1 to 3 150), validation (15 %, records 3 151 to 3 825), and test (15 %, records 3 826 to 4 500) partitions. Hyperparameters were selected on the validation partition, and the test partition was used only for the final evaluation. Inputs were Z-score normalized. The input consisted of sliding windows of 30 consecutive values of σ_a^2 (one per second), and the target was the next value.

Temperature measured by the LM35 was recorded for parallel monitoring but was not used as a model input: the thermal time constant of the motors exceeds the duration of the degradation experiments, so temperature carries little information on the time scale being forecast.

Two architectures were evaluated, single-layer and two-layer stacked LSTM. A partial search of 11 configurations explored unit counts (16 to 128), epochs (800 and 1600), and batch sizes (16, 32, and 64); all results are reported in Table II. All models used the RMSprop optimizer and mean squared error (MSE) loss. Each configuration was trained once.

Forecast fidelity is quantified with a similarity score S, which compares the summary distribution statistics of the predicted and actual series over an evaluation window:

$$S = 100\left(1 - \frac{1}{3}\sum_{\theta\in\{\mu,\,\sigma^2,\,\sigma\}}\frac{|\hat{\theta}-\theta|}{|\theta|}\right)\ \%, \qquad (1)$$

where $\theta \in \{\mu, \sigma^2, \sigma\}$ denotes the mean, variance, and standard deviation of the observed series, and $\hat{\theta}$ the corresponding statistic of the predicted series. Because S measures distributional similarity rather than point-wise agreement, it is complemented with the root mean squared error (RMSE), mean absolute error (MAE), coefficient of determination (R^2), and Pearson correlation (r).

Long-range forecasting used an autoregressive procedure: the trained model was initialized with 1 500 real records and then iteratively predicted the next time step using its own previous output, up to 21 500 steps (about 6 h).

IV. Experimental Setup

The experiments were conducted in 2020 as a proof of concept [12].. Validation used two AC induction motors (Fig. 2): a single-phase capacitor-start motor (220 V, 550 W, 4.7 A, 2 840 r/min, 50 Hz, IP44) and a four-pole three-phase motor (Kohlbach 90S, 380 V, 1.1 kW, 1 450 r/min, 50 Hz). A single sensing node was used, mounted on one motor at a time. Two complementary experimental protocols were applied.

TABLE I
Open-Source Software Stack

Component	Tool	License
Firmware	Arduino + ESP8266 libs	LGPL / Apache 2.0
Database	MariaDB	GPL v2
ML framework	TensorFlow / Keras	Apache 2.0
PCB design	KiCad 6.0	GNU GPL v3
Dashboard	Grafana / InfluxDB	AGPL / MIT
OS (gateway)	Raspberry Pi OS Lite	GPL (Debian)

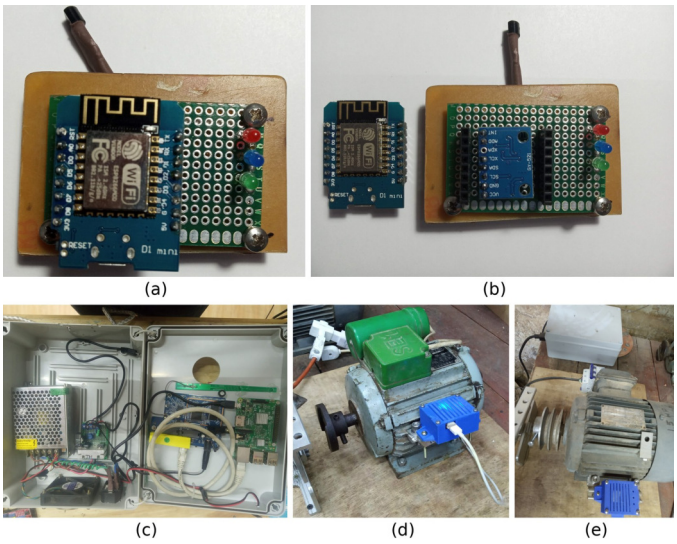


Fig. 2. Experimental hardware: (a) sensing node; (b) node without the D1 Mini, showing the GY-521 (MPU-6050) breakout; (c) gateway enclosure (power supply, Wi-Fi router, Raspberry Pi 3B); (d) node on the single-phase motor; (e) node on the three-phase motor.

Protocol 1 — Static imbalance: four calibrated masses (13.39 g, 16.46 g, 32.92 g, and 43.37 g) were sequentially attached to the motor shaft at a fixed radius of 25 mm. Five 9 min acquisition sessions were recorded per mass level, beginning with the unloaded shaft as baseline. This protocol validated the ability of the sensor to detect discrete vibration changes.

Protocol 2 — Progressive loading: a custom lever mechanism applied a continuously increasing radial load to the motor rotor without stopping the motor. The lever carried a fixed counterweight (2.75 kg in Experiment 1; 3.55 kg in Experiment 2) and a 5 L water reservoir that gradually emptied by drip, increasing the effective lever force over time. This produced a monotonically increasing vibration signal; each run was stopped once the gateway database had received 4 500 time-stamped records (one per second, approximately 75 min). This protocol emulates the monotonic evolution of gradual, non-catastrophic mechanical degradation; it does not reproduce a specific fault mechanism such as bearing wear.

The alarm thresholds were taken as 1.8 mm/s RMS (Zone A/B) and 4.5 mm/s RMS (Zone B/C). Both values lie within the typical zone-boundary ranges given in Annex B of ISO 20816-1 [2]; in the absence of machine-specific limits, they were adopted as reference thresholds for both test motors.

TABLE II
LSTM HYPERPARAMETER SEARCH RESULTS (TOP THREE IN BOLD)

Test	Layers	Units	Batch	Epochs	Time (s)	S (%)
1	2	64/32	32	800	856	68.53
2	2	128/64	32	800	996	64.16
3	2	128/64	64	800	885	64.66
4	2	64/16	32	800	749	87.12
5	**1**	**64**	**16**	**800**	**829**	**91.93**
6	**1**	**64**	**32**	**800**	**741**	**90.41**
7	1	64	64	800	818	71.88
8	1	64	16	1600	1080	76.34
9	1	64	32	1600	1131	89.22
10	1	64	64	1600	1095	89.22
11	**1**	**32**	**16**	**800**	**744**	**95.30**

TABLE III
POINT-WISE METRICS ON HELD-OUT DATA (NORMALIZED UNITS)

Model	Subset	RMSE	MAE	R^2	r
Test 6	A	0.785	0.601	-0.891	0.156
Test 5	A	0.456	0.366	0.287	0.771
Test 11	**A**	**0.404**	**0.323**	**0.457**	**0.819**
Test 11	B	0.574	0.430	-0.076	0.589

TABLE IV
SYSTEM PERFORMANCE SUMMARY

Metric	Value
Min. detected imbalance mass	13.39 g (+35.6 %)
Similarity score S (Test 11)	95.30 %
r / R^2 (Test 11, subset A)	0.819 / 0.457
r / R^2 (Test 11, subset B)	0.589 / -0.076
Model parameters	4 385
Training time	12 min 24 s
Zone A/B crossing (indicative)	≈1 h 32 min after seed
Forecast horizon	≈6 h (projection)
Hardware BOM per node	$<$ USD 15
Software licensing	100 % open-source

V. RESULTS

A. Sensor Validation — Static Imbalance

The MPU-6050 detected all four imbalance levels with a consistent monotonic response on both motor types when evaluated as relative change over the baseline. The smallest mass (13.39 g) produced a 35.6 % increase in X-axis acceleration standard deviation over the unloaded baseline, and the largest (43.37 g) produced an increase of 280.3 %. Although the centrifugal force of a given imbalance scales with ω^2, so that absolute vibration levels are higher on the 2 840 r/min single-phase motor than on the 1 450 r/min three-phase motor, the baseline-normalized relative increases showed close agreement between both machines.

B. LSTM Hyperparameter Search

Table II lists the 11 evaluated configurations. Three of them exceeded $S = 90\,\%$. The best configuration (Test 11: one layer, 32 units, batch size 16, 800 epochs) achieved $S = 95.30\,\%$ with the shortest training time among the top three (744 s). Two-layer architectures and training for 1600 epochs yielded lower S in this search. Since each configuration was trained once, small differences between the top configurations should be interpreted with caution.

C. Point-Wise Forecasting Accuracy

Table III reports point-wise metrics computed on held-out test data, which were not used for training or hyperparameter selection. These metrics come from the original 2020 evaluation logs; since the raw datasets are no longer available, the exact composition of the evaluation subsets could not be reconstructed. Rows 1–3 share the same evaluation subset (subset A), and row 4 corresponds to a larger subset of held-out data, three times the size of subset A (subset B). On subset A, Test 11 achieved RMSE $= 0.404$, MAE $= 0.323$, $R^2 = 0.457$, and $r = 0.819$, outperforming Test 5 ($R^2 = 0.287$) and Test 6 ($R^2 = -0.891$), consistently with the ranking given by S. Notably, Test 6 obtained $S = 90.41\,\%$ despite a negative R^2, illustrating that S captures distributional similarity but not trajectory agreement; point-wise metrics are therefore reported alongside it. On subset B, Test 11 obtained RMSE $= 0.574$ and MAE $= 0.430$, but $R^2 = -0.076$ and $r = 0.589$: on this larger subset, point-wise accuracy was not better than a constant-mean predictor. The model therefore tracks the degradation trend on part of the held-out data, but its point-wise accuracy does not hold over the larger subset, a plausible consequence of the extrapolation inherent to a chronological split of a monotonic signal, and consistent with the proof-of-concept scope of this study.

D. Long-Range Forecasting Against ISO 20816-1 Thresholds

The selected model was applied autoregressively to Experiment 2, an independent dataset with a lever counterweight of 3.55 kg versus 2.75 kg in training. Starting from 1 500 real seed records (25 min), the model generated 21 500 predictions covering approximately 6 h. The forecast indicator crossed the Zone A/B threshold (1.8 mm/s RMS) approximately 1 h 32 min after the end of the seed sequence, and the Zone B/C threshold (4.5 mm/s RMS) approximately 3 h 36 min after it. In the original analysis, these crossing times were obtained from a heuristic composite indicator (mean plus variance of the forecast series over successive windows) compared against the velocity thresholds; the exact mapping between the forecast acceleration variance and vibration velocity could not be reconstructed from the available records. The reported times should therefore be read as indicative of the forecast trend, not as calibrated threshold-crossing predictions. A consistent mapping, for instance the single-harmonic relation $\hat{v}_{\mathrm{RMS}} = \sqrt{\hat{\sigma}_a^2}/(2\pi f_1)$, where f_1 is the rotational frequency, or forecasting v_{RMS} directly, is left for future work. Because Experiment 2 lasts about 75 min, both crossings fall beyond the observed record; they are model projections, not verified events.

VI. DISCUSSION

A lightweight single-layer LSTM (4 385 parameters) reached $S = 95.30\,\%$ and $r = 0.82$ on a held-out subset,

outperforming deeper two-layer configurations in this search, although its point-wise accuracy did not hold on a larger held-out subset. Accuracy values reported in related work, such as 93 % in [9] and above 92 % in [10], refer to different datasets, tasks (classification rather than forecasting), and metrics, and are therefore not directly comparable with S. The relevant difference lies elsewhere: the present system achieves its results with commodity hardware below USD 15 per node and a fully open stack. The result suggests that a compact recurrent model can capture structured, monotonic degradation trends, although confirming this would require repeated training runs and longer evaluation horizons.

The projected Zone A/B crossing approximately 1 h 32 min ahead indicates that, if confirmed on longer real records, the forecast could support scheduling interventions during natural production transitions, such as shift changes or batch changeovers, reducing both unscheduled downtime and premature component replacement.

The reported cost corresponds to the bill of materials and PCB fabrication only. A commercial product additionally includes industrial-grade enclosures, electromagnetic compatibility and ingress-protection certification, installation, documentation, warranty, technical support, and the manufacturer's margin; these components are not accounted for here, so the figure should not be read as a direct price comparison with commercial solutions such as those of WEG or ABB. The value of the proposed system lies rather in data ownership, the absence of subscription fees, unrestricted access to historical data, and full modifiability, which make it a reproducible baseline for research and small-scale industrial applications.

Several limitations should be acknowledged. (i) The progressive loading protocol emulates monotonic, non-catastrophic degradation in a controlled laboratory setting with two motors; it does not cover the stochastic diversity of real industrial faults. (ii) The evaluated vibration band is limited to 10 Hz to 260 Hz rather than the 10 Hz to 1 000 Hz band of ISO 20816-1, and the acceleration-to-velocity conversion was not validated against a reference vibrometer. (iii) The I^2C bus of the MPU-6050 may be susceptible to electromagnetic interference in high-noise environments. (iv) S is a domain-specific metric that does not reflect trajectory agreement; point-wise metrics show that accuracy on a larger held-out subset was limited ($R^2 = -0.076$), and each configuration was trained only once, so run-to-run variability was not quantified. (v) Long-range crossings are projections beyond the observed record, obtained with a heuristic acceleration-to-velocity mapping.

VII. Conclusion

This paper presented the design and proof-of-concept laboratory validation of a fully open-source, low-cost predictive maintenance system for AC induction motors. The results show: (1) detection of mechanical imbalance from 13.39 g on both test motors; (2) a lightweight 4 385-parameter LSTM that reaches $S = 95.30\,\%$ and $r = 0.82$ on a held-out subset; (3) point-wise metrics that reveal limited accuracy on a larger held-out subset ($R^2 = -0.076$), motivating the hardware and validation improvements below; and (4) an autoregressive forecast that projects the crossing of the ISO 20816-1-referenced Zone A/B threshold approximately 1 h 32 min ahead. The work contributes a complete, reproducible end-to-end framework combining low-cost hardware with open-source machine learning for motor condition monitoring.

Future work includes migrating the node to an ESP32 with an ADXL355 accelerometer to cover the full 10 Hz to 1 000 Hz ISO 20816-1 band and improve noise immunity, validating the velocity chain against a reference vibrometer, quantifying run-to-run variability with repeated training, and benchmarking against persistence and linear-extrapolation baselines. Architectural alternatives such as Temporal Convolutional Networks, Temporal Fusion Transformers, and Graph Neural Networks for multi-asset modeling will also be explored. Final validation will target TRL 7 deployment in an operational agro-industrial plant.

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